ACADEMIC RESEARCH & IMPLEMENTATION

AI-Generated Face Detection

A deep learning approach for facial authenticity verification leveraging pre-trained MobileNetV3-Large, RetinaFace alignment, and advanced regularization techniques.

ARCHITECTURE	FRAMEWORK	INPUT RESOLUTION	BEST PERFORMANCE
MobileNetV3-Large	**PyTorch**	**224 × 224 × 3**	**99.06% Test Acc**

SECTION 1.1 · OVERVIEW

Dataset Description & Preprocessing

1. Dataset Representation & Quality

A deep learning model's face detection effectiveness is largely determined by training data diversity, size, and representation.

2. Authentic + Synthetic Data

A robust classifier must combine authentic human face images with synthetic images produced by state-of-the-art generators.

3. Multi-Generator Training

Training on multiple generators ensures the neural network uncovers general facial irregularities instead of generator-specific patterns.

4. Real + Synthetic Combination

This study incorporates real human images (FFHQ) and synthetic faces from two separate generators (Stable Diffusion & Nano Banana 2.0).

Dataset Overview

Total Images	**~120K+**
Real Images (FFHQ)	**~40K+**
Synthetic (Stable Diffusion)	**~40K+**
Synthetic (Nano Banana 2.0)	**~40K+**
Image Resolution	**224 × 224**
Train / Val / Test Split	**70% / 15% / 15%**

Sources

FFHQ

Stable Diffusion

Nano Banana 2.0

SECTION 1.2 · DATASETS BREAKDOWN

Datasets Utilized

1. Real Faces (FFHQ)

- Source: Flickr-Faces-HQ dataset
- Popular benchmark for face verification
- Diverse in age, gender, ethnicity, pose, background
- Reduces dataset bias

15,000

real images selected

2. Stable Diffusion

- Modern diffusion-based generation
- Higher realism than early GAN networks
- Challenges model with subtle synthetic irregularities
- Forces detection of lighting, geometry, noise cues

9,001

AI-generated images

3. Nano Banana 2.0

- Evaluates cross-domain generalization
- Overcomes dependency on Stable Diffusion anomalies
- Merged into train, validation, and test splits
- Expands general face characteristics

31,999 / 4,000 / 4,000

train / val / test

SECTION 1.3 · BINARY SPLITS

Dataset Partitioning & Construction

- Target labels: Class 0 (Real Human Portrait) & Class 1 (AI-Generated Face)
- Initial stratified 80:10:10 split, later merged with Nano Banana 2.0 partitions

Primary Dataset Construction

Category	Number of Images
Real Images (FFHQ)	15,000
AI-generated Images (Stable Diffusion)	9,001
Primary Total	24,001

Final Consolidated Dataset

Dataset Split	Number of Images
Training Split	51,199
Validation Split	6,400
Testing Split	6,401
Final Combined Total	64,000

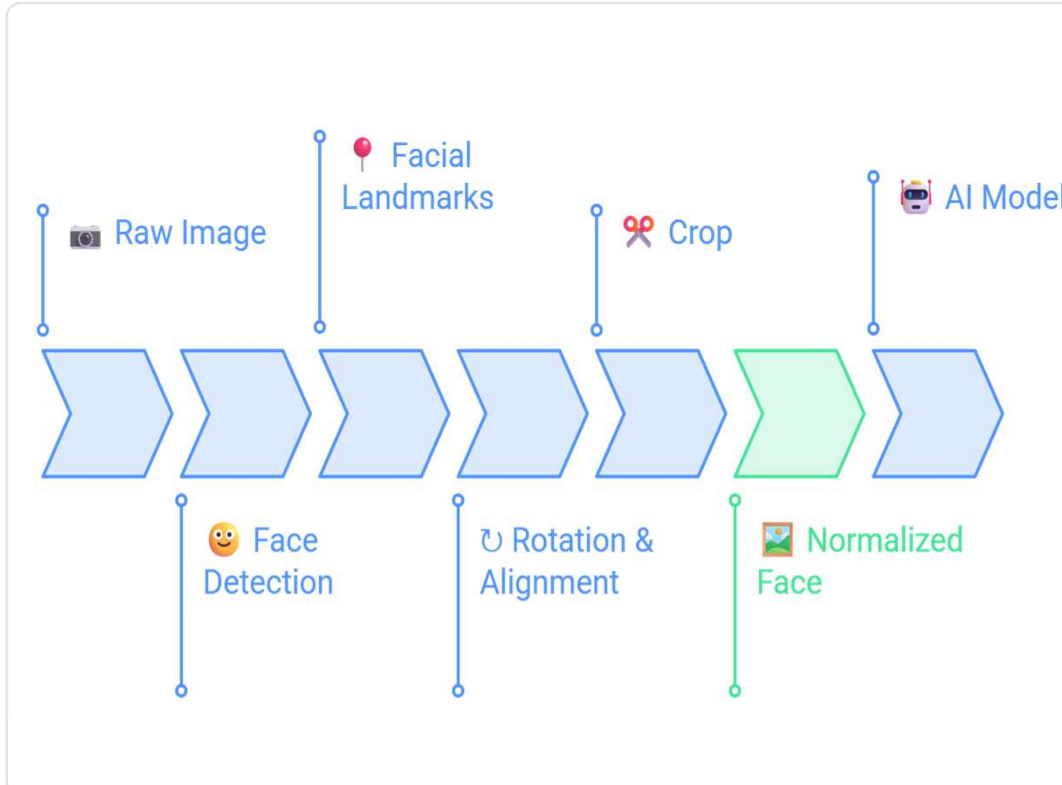
64,000 total images across the final combined dataset, spanning one real-image source (FFHQ) and two independent AI generators (Stable Diffusion, Nano Banana 2.0) to reduce generator-specific bias.

SECTION 1.4 · FACE ALIGNMENT & NORMALIZATION

Data Preprocessing Workflow

RetinaFace Alignment & Normalization

- Raw images vary in head pose, angle, illumination, background detail, and quality
- A multi-stage pipeline removes variables unrelated to face authenticity
- RetinaFace extracts face coordinates and maps 5 facial landmarks (eyes, nose, mouth corners)
- Landmark alignment rotates and scales key details to static spatial positions
- Reduces background bias and head tilt, forcing the classifier to study texture
- Fallback center-crop is applied if the detector fails to identify facial coordinates



SECTION 1.4 · PIPELINE FLOW

Preprocessing Pipeline Stages

Complete sequential pipeline mapping from Raw Input to Processed Output:

1 Face Detection

RetinaFace locates the face and filters noisy pixels.

2 Landmark Alignment

Rotates coordinates based on eyes and nose corners.

3 Face Cropping

Removes backgrounds, hair, and clothing shapes.

4 Spatial Resizing

Resizes to 224×224 for MobileNetV3 compatibility.

5 Mean Normalization

Normalizes RGB channels using ImageNet parameters.

6 Data Augmentation

On-the-fly transforms applied to training subsets only.

SECTION 1.4.2 & 1.4.3

Face Cropping, Resizing & Normalization

Spatial Resolution and Scale

- Crops the aligned facial box strictly to remove background details
- Landscapes, illumination sources, or outfits introduce training bias
- Resizes crops to standard 224 × 224 dimensions
- Preserves vital facial textures, iris shapes, and boundary outlines
- Ensures uniform grid sizing for fast batch compilation and low GPU memory overhead

ImageNet Input Normalization

Ensures input pixels follow pre-trained weight distributions.

Parameter	Red	Green	Blue
Mean (ImageNet)	0.485	0.456	0.406
Std Dev (ImageNet)	0.229	0.224	0.225

Advantages of standardization

- Stabilizes model weight adjustments during early training
- Ensures compatibility with ImageNet features, improving adaptation speed

On-the-Fly Data Augmentation

Geometric Transforms

- Random Resized Crop — extracts different regions of the facial image, improving robustness to scale and framing
- Horizontal Flip — flips images with 50% probability, developing left-right invariance

Color & Blur

- Color Jitter — modifies brightness, contrast, saturation, and hue to simulate lighting/camera variance
- Gaussian Blur — blurs details with variable kernels, simulating focal errors or motion blur

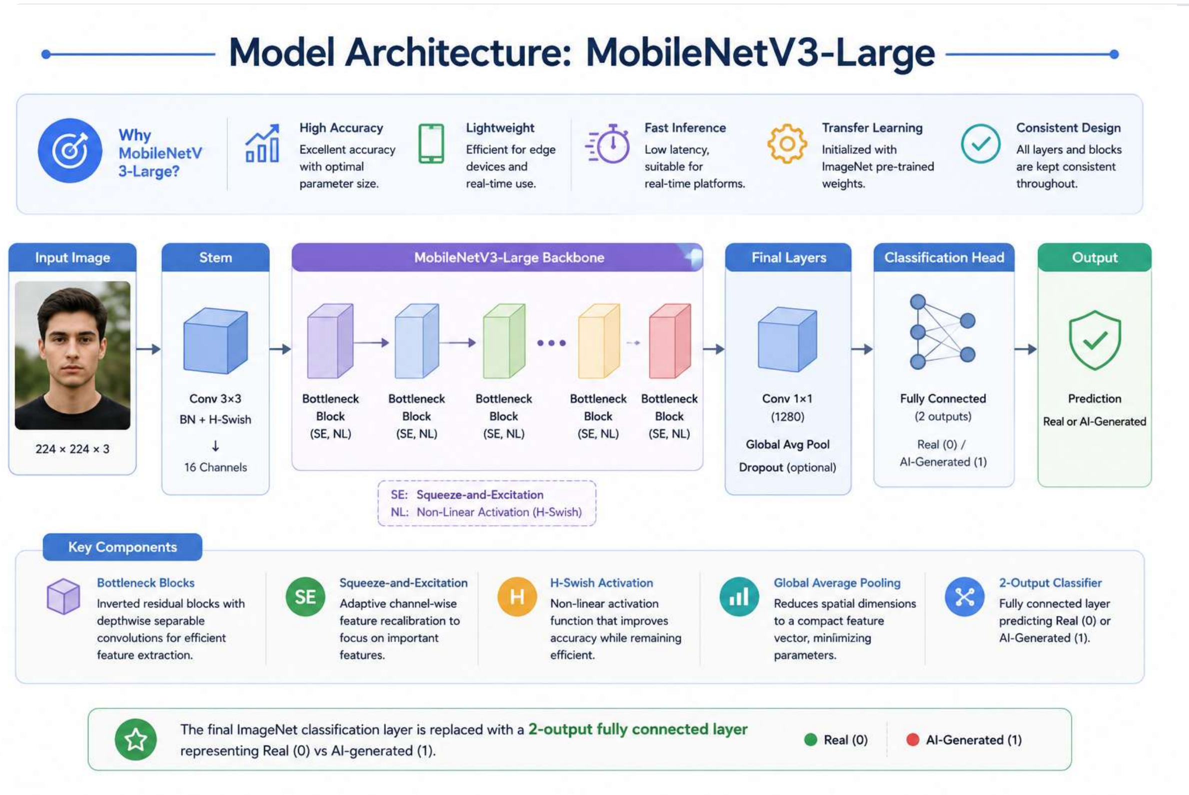
Quality Degradations

- JPEG Compression — prepares the model to identify deepfakes on compressed social platforms
- Gaussian Noise — perturbs pixel values, forcing focus on structure over individual pixels

SECTION 2.1 & 2.2 · NEURAL LAYOUT

Model Architecture: MobileNetV3-Large

- Utilizes Transfer Learning with the MobileNetV3-Large backbone, initialized with ImageNet pre-trained weights
- Offers an optimal balance between accuracy, parameter size, and latency
- Lightweight design supports easy deployment on edge devices and real-time platforms
- Final ImageNet classification layer replaced with a 2-output fully connected layer: Real (0) vs AI-Generated (1)
- All intermediate feature extraction blocks and internal layers are kept consistent



SECTION 2.3 · STRATEGY TIMELINE

Two-Stage Transfer Learning

Decoupled training stages prevent catastrophic loss of generic filters:

EPOCHS 1 – 3

Stage 1: Feature Extraction

All parameters in the MobileNetV3-Large feature extraction backbone are frozen. Only the new classification layer is trained.

3×10^{-4}

LEARNING RATE

Allows the classifier to learn high-level face shapes while preserving generic pre-trained visual representations.

EPOCHS 4 – 10

Stage 2: Selective Fine-Tuning

The final 25% of MobileNetV3-Large feature blocks (blocks 12–16) are unfrozen and optimized.

1×10^{-5}

LEARNING RATE

Adapts higher-level layers to identify subtle AI artifacts without destroying low-level edge filters.

SECTION 3.1 & 3.2 · SETTINGS

Training Configuration & Environment

Runtime Environment Details

Component	Specification
Platform	Kaggle Notebooks
Framework	PyTorch
Computer Vision Library	Torchvision
Hardware Accelerator	Kaggle GPU Runtime
Mixed Precision Training	Automatic Mixed Precision (AMP)

Dataloader Parameters

Parameter	Value
Input Image Size	224 × 224 pixels
Number of Channels	3 (RGB)
Batch Size	128
Number of Classes	2 (Real, AI-Generated)
Data Shuffling	Enabled for Training

SECTION 3.4 – 3.6

Optimization Details

1. Loss Function

- Uses standard CrossEntropyLoss
- Directly compares predicted logits with ground truth binary classes
- Provides stable gradients when paired with softmax outputs

2. Optimizer (AdamW)

- AdamW selected over basic Adam
- Decouples weight decay from the gradient update step
- Weight Decay of 1×10^{-4}; Stage 1 LR 3×10^{-4}, Stage 2 LR 1×10^{-5}

3. Cosine Scheduler

- Uses CosineAnnealingLR scheduler
- Decreases learning rate smoothly following a half-cosine wave
- Enables large updates early, minor adjustments near the optimum

SECTION 4.1 & 4.2 · ABLATIONS

Hyperparameter Optimization Study

Methodology & Setup

- Conducted a systematic study with 24 controlled ablation experiments
- A single parameter is altered per experiment, keeping all others static
- Aims to optimize accuracy and F1-score while avoiding overfitting
- Evaluates learning rate, batch size, regularization decay, dropout, optimization method, and label smoothing

Evaluated Parameter Ranges

Hyperparameter	Tested Values
Learning Rate	1e-4, 5e-4, 1e-3, 5e-3
Batch Size	32, 64, 128
Weight Decay	0.0, 0.01, 0.05, 0.10
Dropout Rate	0.0, 0.2, 0.3, 0.5
Optimizer	Adam, AdamW
LR Scheduler	CosineAnnealingLR, ReduceLROnPlateau
Label Smoothing	0.0, 0.05, 0.10

Ablation: Learning Rate & Batch Size

Learning Rate (LR) Performance

Rates of 5×10^{-4} and 1×10^{-3} achieved optimal accuracy. High rates caused divergence; low rates slowed adaptation.

Accuracy	Learning Rate	
92.71%	1×10^{-4}	
95.43%	5×10^{-4}	SELECTED
95.41%	1×10^{-3}	
90.10%	5×10^{-3}	

Batch Size Performance

Batch size 32 offered marginal gains, but 128 was selected to maximize GPU acceleration.

Accuracy	Batch Size	
95.57%	Batch Size 32	
95.41%	Batch Size 64	
95.51%	Batch Size 128	SELECTED

SECTION 4.5 – 4.7

Ablation: Regularization & Optimizer

Dropout & Weight Decay

No additional dropout (0.0) worked best. Weight decay (0.05–0.10) improved F1 stability.

Adam vs AdamW Optimizer

AdamW's decoupled decay resulted in a 9% test accuracy improvement over standard Adam.

95.73%
Dropout 0.0
SELECTED

95.41%
Dropout 0.2

95.54%
Weight Decay 0.05

95.57%
Weight Decay 0.10

86.51%
Standard Adam

95.41%
AdamW Optimizer
SELECTED

SECTION 4.8 & 4.9

Ablation: Scheduler & Label Smoothing

Learning Rate Scheduler

Cosine decay outperformed plateau-based reductions by adapting parameters more smoothly.

95.41%	CosineAnnealing	SELECTED
93.23%	ReduceLROnPlateau	

Label Smoothing

Smoothing did not offer benefits, as clean targets and dataset size minimized label noise.

95.41%	No Smoothing (0.0)	SELECTED
93.10%	Smoothing 0.05	
94.30%	Smoothing 0.10	

Generalization & Training Stability

SECTION 6.1 & 6.2 · MODEL PERFORMANCE

Quantitative Results & Classification Report

The optimized MobileNetV3-Large model achieved excellent classification performance, with minimal generalization gap between validation and test sets.

99.30%

BEST VALIDATION ACCURACY

99.06%

FINAL TEST ACCURACY

Classification Report

Class Label	Precision	Recall	F1-Score
Real Faces	0.9900	0.9929	0.9914
AI-Generated Faces	0.9914	0.9879	0.9896

Report Insights

- High precision and recall for both classes show balanced predictions
- A low generalization gap verifies the dataset was shuffled and split correctly

SECTION 7 · CLOSING

Summary Conclusions

Key Findings

- High Accuracy — reached a test accuracy of 99.06% and validation accuracy of 99.30%
- Model generalizability — cross-domain training (Nano Banana 2.0) successfully reduced generator bias
- Efficient architecture — MobileNetV3-Large proved highly computationally efficient

Future Work

- Test on newer models like Stable Diffusion XL or Midjourney v6
- Incorporate frequency-domain analysis to detect subtle synthetic textures
- Utilize explainable AI (Grad-CAM) to verify classification features